

PRODUCT SPECIFICATION

Product Name	S76S LoRa™ Wireless Communication Module
Version	L
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An IoT Solution Company

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Document History

Date	Revised Contents	Revised By	Version
July 20 th , 2016	Draft Version	Chunyi	A
Aug 25 th , 2016	Revised 7.1 marking without logo	Nick	B
Sep 10 th , 2016	Revised marking with Acsip logo	Nick	C
Sep 26 th , 2016	Revised 4.1 footprint drawing	Nick	D
Oct. 7 th , 2016	Update footprint	Nick	E
Oct. 11 th , 2016	Add pin assignment, revise footprint and mechanical dimension	Kenny	F
Mar 22, 2017	Modify Marking information	Kenny	G
Apr 10 th , 2017	Operation temp. range modified	Chunyi	H
May 22 nd , 2018	RF Characteristics and pin definition updated	Kenny	I
July 08 th , 2019	Modify Marking information. Add 2D Barcode	Jack	J
Aug. 14 th , 2019	Modify Marking information. Delete 2D Barcode	Jack	K
Feb 17, 2020	Modify Packing Information Pin1 Location Add Temp. Note@ SF=12(Page.3)	PW	L



1. General Description

The S76S integrates ARM Cortex®-M0+ (32-bit RISC core operating at a 32MHz frequency) MCU with LoRa™ modulation that provides ultra-long range spread spectrum communication and high interference immunity whilst minimizing current consumption.

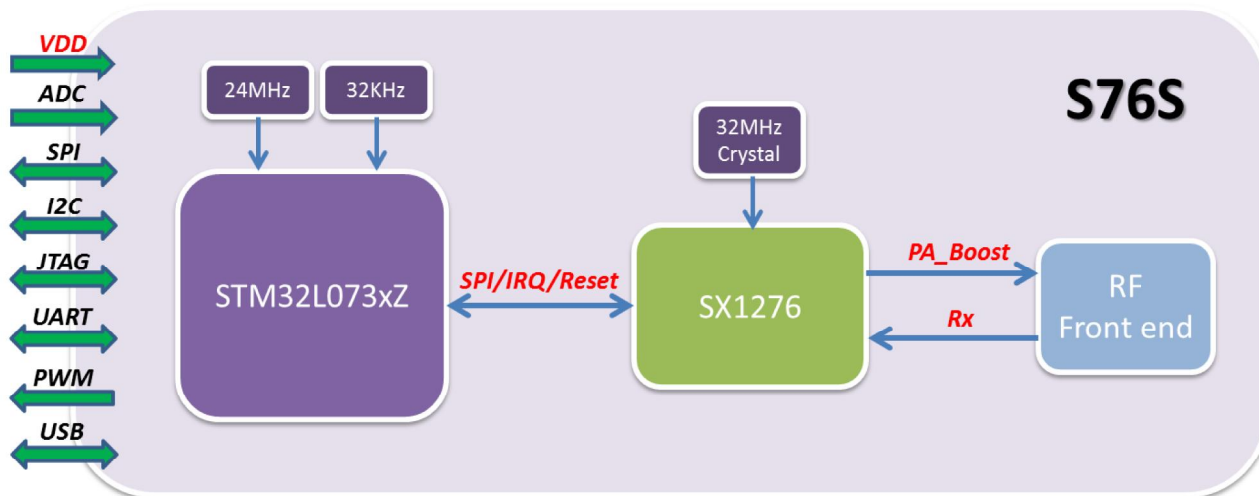
S76S can achieve a sensitivity of over -137 dBm. The high sensitivity combined with the integrated +20 dBm power amplifier yields industry leading link budget making it optimal for any low data rate application requiring range or robustness. LoRa™ also provides significant advantages in both blocking and selectivity over conventional modulation techniques, solving the traditional design compromise between range, interference immunity and energy consumption.

Feature

- Small footprint : 13 mm x 11 mm x 1.1 mm
- LoRa™ Modem
- +20 dBm constant RF output vs. V supply
- Programmable bit rate up to 37500 bps
- High sensitivity: down to -137 dBm
- Excellent blocking immunity
- Preamble detection
- Automatic RF Sense and CAD with ultra-fast AFC
- Payload up to 128 bytes with CRC
- Embedded memories (up to 192 Kbytes of Flash memory and 20 Kbytes of RAM)

1-1 Block Diagram

A simplified block diagram of the S76S module is depicted in the figure below.



1-2 Product Version

The features of S76S is detailed in the following table

Part Number	Frequency Range	Spreading Factor	Bandwidth (KHz)	Effective Bitrate (bps)	Est. Sensitivity (dBm)
S76S	920-925 MHz(AS923)	6 - 12**	62.5 - 500	146 - 37500	-109 to -137*

Note: * LORA setting SF=12, BW=62.5k, Long-Range Mode, highest LNA gain, *LnaBoost* for Band 1.

** LORA setting SF=12, suggest to add Offset on TX freq. to compensate temperature
Coefficient of XTAL@ Temp < -20°C

1-3 Specification

Model Name	S76S
Product Description	LoRa Wireless Communication Module
Host Interface	UART
Operation Conditions	
Temperature	- Storage : -50°C ~ +105°C - Operating : -40°C ~ +85°C
Humidity	- Operating : 10 ~ 95% (Non-Condensing) - Storage : 5 ~ 95% (Non-Condensing)
Dimension	13 mm x 11 mm x 1.1 mm
Package	LGA type

2. Electrical Characteristics

2-1. Absolute Maximum Ratings

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDD33	Supply Voltage	-0.3		3.9	V
V _{IN}	Input voltage on digital pins	-0.3		3.9	V
P _{mr}	RF Input Level			+10	dBm

2-2. Recommended Operating Range

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDD33	Supply Voltage	2.4	3.3	3.6	V
ML	RF Input Level			+10	dBm

2-3. Power Consumption Characteristics

Symbol	Parameter	Conditions	Typ.	Max.	Unit
IDDSL	Supply current in Sleep mode	Sleep Stop Mode		5	uA
IDDST	Supply current in Standby mode	Crystal oscillator enabled	9	9.6	mA
IDDR	Supply current in Receive mode		17.5		mA
IDDT	Supply current in Transmit mode with impedance matching	RF SetPW = +20 dBm RF SetPW = +17 dBm RF SetPW = +13 dBm RF SetPW = + 7 dBm	127 82 65 49		mA

2-4. RF Characteristics

The table below gives the electrical specifications for the transceiver operating with LoRa™ modulation.

Following conditions apply unless otherwise specified:

- Supply voltage = 3.3 V.
- Temperature = 25° C.
- Frequency bands: 920~925MHz
- Bandwidth (BW) = 125 kHz.
- Spreading Factor (SF) = 12.
- Error Correction Code (EC) = 4/6.
- Packet Error Rate (PER)= 1%
- CRC on payload enabled.
- Output power = 13 dBm in transmission.
- Payload length = 64 bytes.
- Preamble Length = 12 symbols (programmed register Preamble Length=8)
- With matched impedances

LoRa Transmitter (Conductive)

Item	Condition	Min.	Typ.	Max.	Unit
Frequency Range	Band1	920	923	925	MHz
Tx Pwr Level @Module O/P	PA_BOOST pin	17.5	18.5	20.0	dBm

LoRa Receiver (Conductive)

Item	Condition	Min.	Typ.	Max.	Unit
Frequency Range	Band1	920	923	925	MHz
RF sensitivity, (Long-Range Mode, highest LNA gain, LNA boost, 62.5 kHz bandwidth)	SF = 10		-133		dBm
	SF = 11		-135		dBm
	SF = 12		-137		dBm
RF sensitivity, (Long-Range Mode, highest LNA gain, LNA boost, 125 kHz bandwidth)	SF = 7		-121		dBm
	SF = 8		-124		dBm
	SF = 9		-127		dBm
	SF = 10		-130		dBm
	SF = 11		-131		dBm
	SF = 12		-134		dBm
RF sensitivity, (Long-Range Mode, highest LNA gain, LNA boost, 500 kHz bandwidth)	SF = 7		-114		dBm
	SF = 8		-117		dBm
	SF = 9		-120		dBm
	SF = 10		-123		dBm
	SF = 11		-126		dBm
	SF = 12		-128		dBm

2-5. Digital Characteristics

2-5-1. DC characteristics

Input voltage levels

Symbol	Description	Conditions	Min	Typ.	Max	Unit
V _{IH}	I/O input high level voltage	NRST	0.7xVDD33	-	-	V
		BOOT0	0.7xVDD33	-	-	V
		GPIO	0.7xVDD33	-	-	V
V _{IL}	I/O input low level voltage	NRST	-	-	0.3xVDD33	V
		BOOT0	-	-	0.14xVDD33	V
		GPIO	-	-	0.3xVDD33	V
R _{PU}	Weak pull-up Equivalent resistor	V _{IN} = GND	30	45	60	K Ω
R _{PD}	Weak pull-down Equivalent resistor	V _{IN} = VDD33	30	45	60	K Ω

Output voltage levels

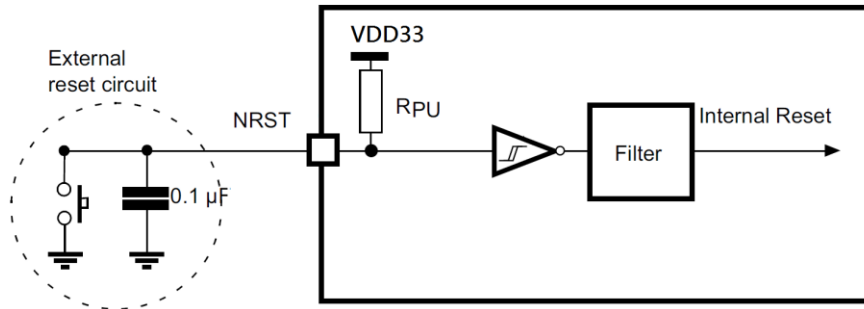
Symbol	Description	Conditions	Min	Max	Unit
V _{OL}	Output low level voltage for an I/O pin	CMOS port / IIO = +8 mA	-	0.4	V
V _{OH}	Output high level voltage for an I/O pin	2.7 V ≤ VDD33 ≤ 3.6 V	VDD33-0.4	-	
V _{OL}	Output low level voltage for an I/O pin	TTL port / IIO = + 8 mA 2.7 V ≤ VDD33 ≤ 3.6 V	-	0.4	
V _{OH}	Output high level voltage for an I/O pin	TTL port / IIO = - 6 mA 2.7 V ≤ VDD33 ≤ 3.6 V	2.4	-	
V _{OL}	Output low level voltage for an I/O pin	IIO = +15 mA 2.7 V ≤ VDD33 ≤ 3.6 V	-	1.3	
V _{OH}	Output high level voltage for an I/O pin	IIO = -15 mA 2.7 V ≤ VDD33 ≤ 3.6 V	VDD33-1.3	-	
V _{OL}	Output low level voltage for an I/O pin	IIO = +4 mA 1.65 V ≤ VDD33 ≤ 3.6 V	-	0.45	
V _{OH}	Output high level voltage for an I/O pin	IIO = +4 mA 1.65 V ≤ VDD33 ≤ 3.6 V	VDD33-0.45	-	

2-5-2. NRST pin characteristics

The NRST pin input driver uses CMOS technology.

It is connected to a permanent pull-up resistor (R_{PU}).

The following figure is recommended NRST pin protection circuit against parasitic resets.



Symbol	Description	Conditions	Min	Typ.	Max	Unit
$V_{IL(NRST)}$	NRST input low level voltage		VSS		0.8	V
$V_{IH(NRST)}$	NRST input high level voltage		1.4		VDD33	V
$V_{OL(NRST)}$	NRST output low level voltage	$I_{OL} = 2\text{mA}$ $2.7\text{V} < \text{VDD33} < 3.6\text{V}$			0.4	V
$V_{OL(NRST)}$	NRST output low level voltage	$I_{OL} = 1.5\text{mA}$ $1.65\text{V} < \text{VDD33} < 2.7\text{V}$			0.4	V
$V_{hys(NRST)}$	NRST schmitt trigger voltage hysteresis			10% VDD33		mV
R_{PU}	Weak pull-up Equivalent resistor	$V_{IN} = \text{GND}$	30	45	60	K Ω
V_F	NRST Input filtered pulse				50	nS
V_{NF}	NRST Input not filtered pulse	$\text{VDD33} > 2.7\text{V}$		350		nS

2-5-3. UART Interface Parameters

Baud Rate = 115200 bps

Data Bits = 8 bits

Stop Bits = 1 bit

Parity Check = None

Flow Control = None

3. Pin Definition

Pin	Definition	I/O	Description
1	NC		
2	GND		Ground pin
3	GND		Ground pin
4	PC0	I/O	MCU pin name: PC0
5	PC1	I/O	MCU pin name: PC1
6	PC2	I/O	MCU pin name: PC2
7	PC3	I/O	MCU pin name: PC3
8	NC		
9	NC		
10	NC		
11	NC		
12	NRST		Hardware reset pin
13	PA0	I/O	MCU pin name: PA0
14	GND		Ground pin
15	GND		Ground pin
16	PA2	I/O	MCU pin name: PA2
17	PA3	I/O	MCU pin name: PA3
18	PA4_SPI1_NSS	I/O	MCU pin name: PA4
19	PA5_SPI1_SCK	I/O	MCU pin name: PA5
20	PA6_SPI1_MISO	I/O	MCU pin name: PA6
21	PA7_SPI1_MOSI	I/O	MCU pin name: PA7
22	PC4	I/O	MCU pin name: PC4
23	PC5	I/O	MCU pin name: PC5
24	PB0	I/O	MCU pin name: PB0
25	PB1	I/O	MCU pin name: PB1
26	PC6	I/O	MCU pin name: PC6
27	PC7	I/O	MCU pin name: PC7
28	PC8	I/O	MCU pin name: PC8
29	PC9	I/O	MCU pin name: PC9
30	RXTX/RFMOD	O	Control signal from SX1276, which connects to internal RF switch at the same time.

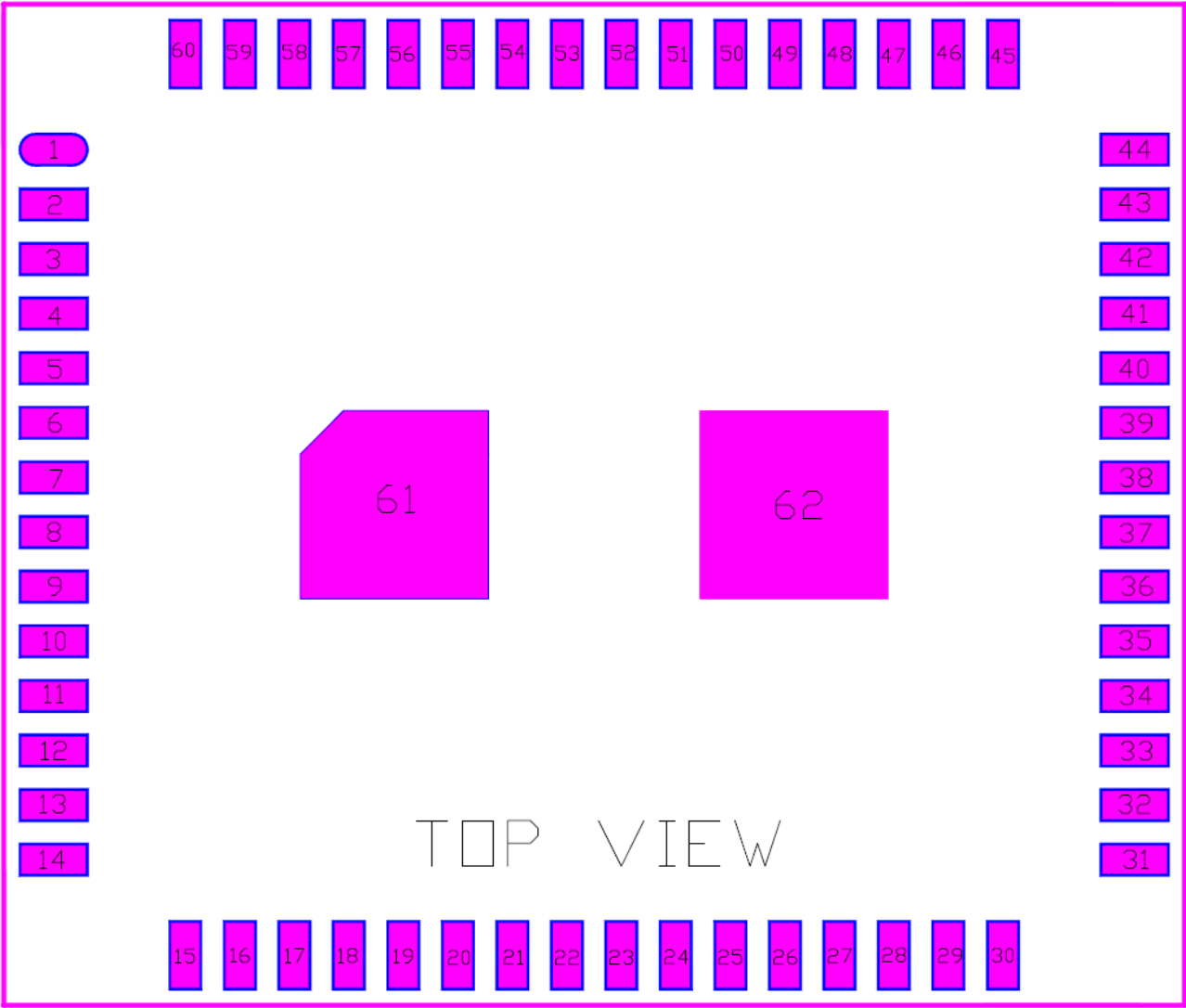
31	GND		Ground pin
32	GND		Ground pin
33	RF_ANT	I/O	RF I/O
34	GND		Ground pin
35	GND		Ground pin
36	PA1_RF_FEM_CPS	O	Control signal from MCU_PA1, which connects to internal RF switch at the same time.
37	GND		Ground pin
38	NC		
39	GND		Ground pin
40	NC		
41	GND		Ground pin
42	NC		
43	VDD33		Power Supply
44	VDD33		Power Supply
45	PA8	I/O	MCU pin name: PA8
46	PA10_USART1_RX	I/O	MCU pin name: PA10
47	PA9_USART1_TX	I/O	MCU pin name: PA9
48	PA11	I/O	MCU pin name: PA11
49	PA12	I/O	MCU pin name: PA12
50	PA13_SWDIO		Serial wire (SWD) debug interface
51	PA14_SWCLK		Serial wire (SWD) debug interface
52	PC10	I/O	MCU pin name: PC10
53	PC11	I/O	MCU pin name: PC11
54	PC12	I/O	MCU pin name: PC12
55	PD2	I/O	MCU pin name: PD2
56	PB5	I/O	MCU pin name: PB5
57	PB6_SCL	I/O	MCU pin name: PB6
58	PB7_SDA	I/O	MCU pin name: PB7
59	BOOT0	I	Boot mode selection pin
60	PB8	I/O	MCU pin name: PB8
61	GND		Ground Pin
62	GND		Ground Pin

※ For detailed functions of pin definitions, please refer to [STM32L073](#) datasheet.



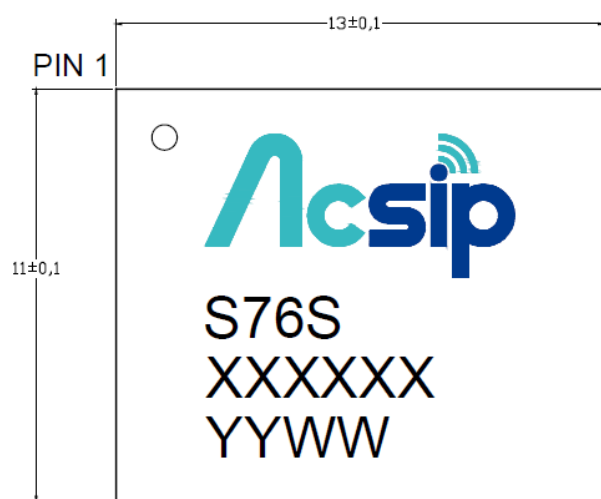
3-1. Pin Assignment

The SiP module will conform to the following pin map, shown in the following diagram (top view)

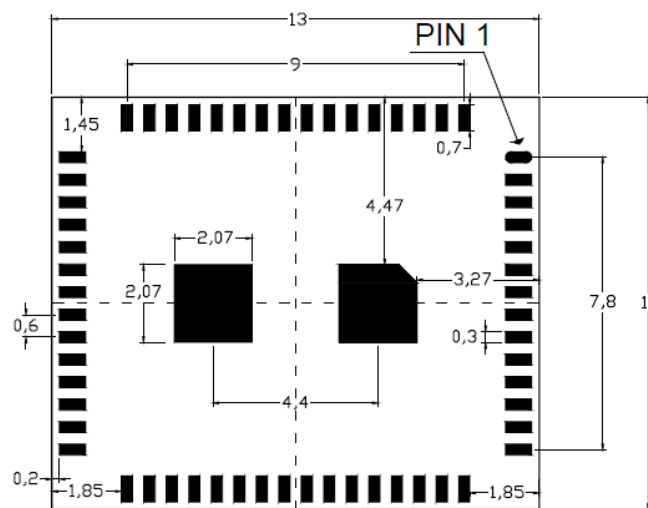


4. Mechanical Dimension

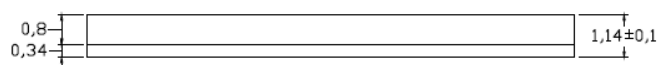
Unit: mm



TOP VIEW



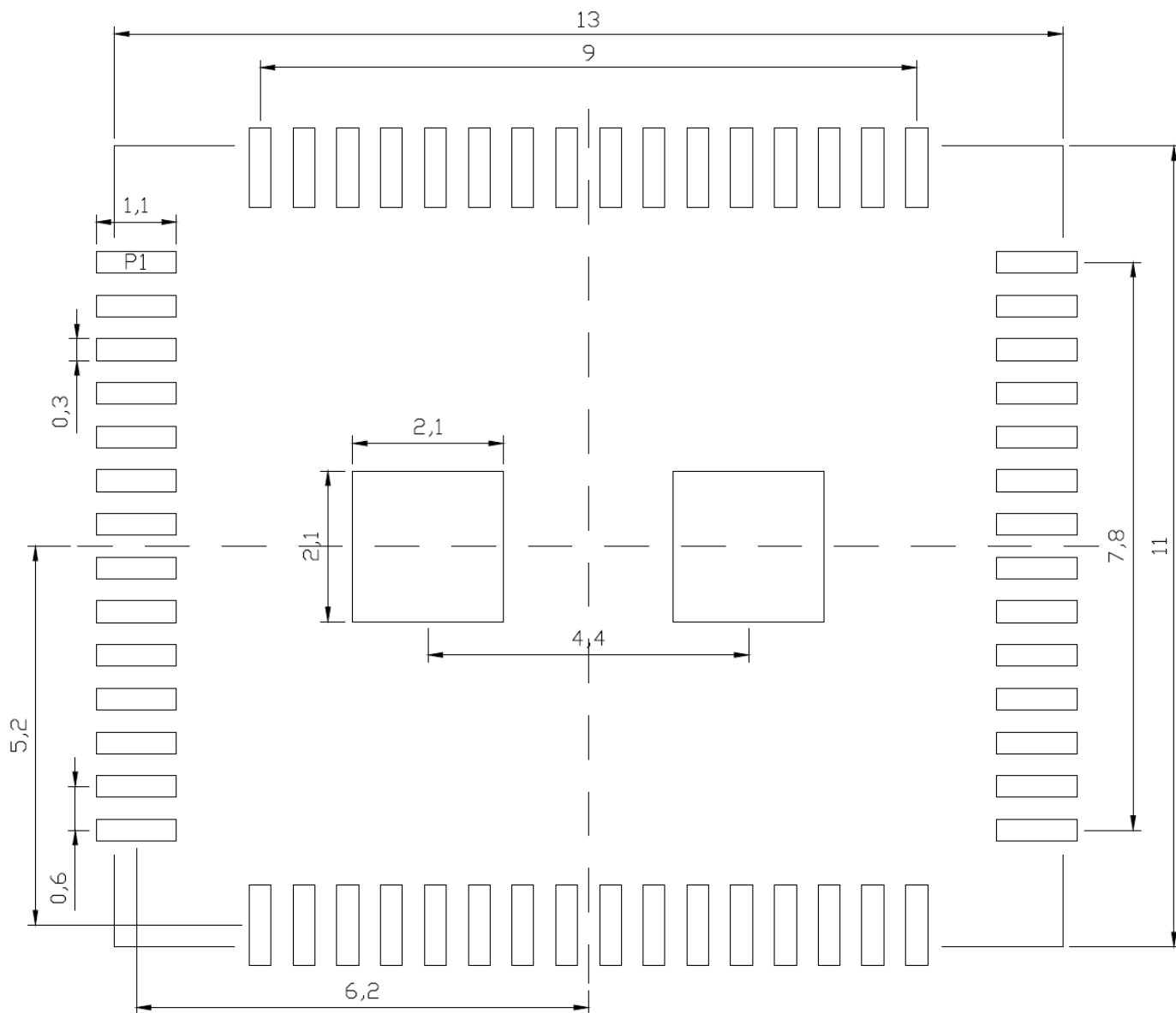
BOTTOM VIEW



SIDE VIEW

4-1 Recommended Footprint

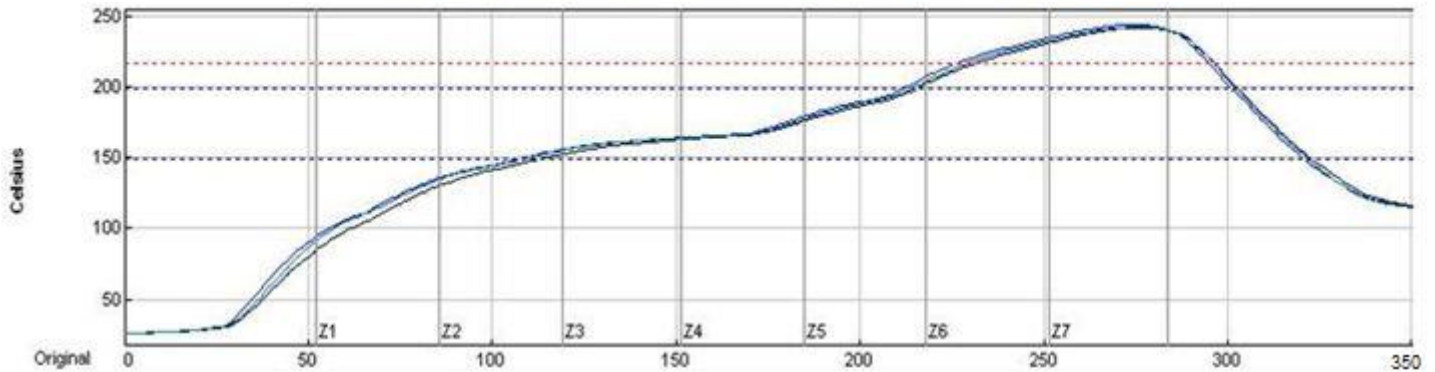
Unit: mm



TOP View

5. Recommended Reflow Profile

Reflow Profile for SiP on board Assembly



Preheat time	150°C—200°C : 105+/-15sec
Dwell time	Over 220°C : 70+5/-10 sec
Peak Temp	240 +10/-5°C
Ramp Up/Down Rate	Up: 3 +0/-2 °C / sec Down: 2 +0/-1°C / sec

6. SiP Module Preparation

6-1. Handling

Handling the module must wear the anti-static wrist strap to avoid ESD damage. After each module is aligned and tested, it should be transport and storage with anti -static tray and packing. This protective package must be remained in suitable environment until the module is assembled and soldered onto the main board.

6-2. SMT Preparation

1. Calculated shelf life in sealed bag: 6 months at <40°C and <90% relative humidity (RH).
2. Peak package body temperature: 250°C.
3. After bag was opened, devices that will be subjected to reflow solder or other high temperature process must.
 - A. Mounted within: 168 hours of factory conditions <30°C /60%RH.
 - B. Stored at $\leq 10\%$ RH with N2 flow box.
4. Devices require baking, before mounting, if:
 - A. Package bag does not keep in vacuumed while first time open.
 - B. Humidity Indicator Card is >10% when read at 23 $\pm$ 5°C.
 - C. Expose at 3A condition over 8 hours or Expose at 3B condition over 24 hours.
5. If baking is required, devices may be baked for 12 hours at 125 $\pm$ 5°C.

7.Package Information

7-1. Product Making

Figure 1 below details the standard product marking for all AcSiP Corp. products. Cross reference to the applicable line number and table for a full detail of all the variables.

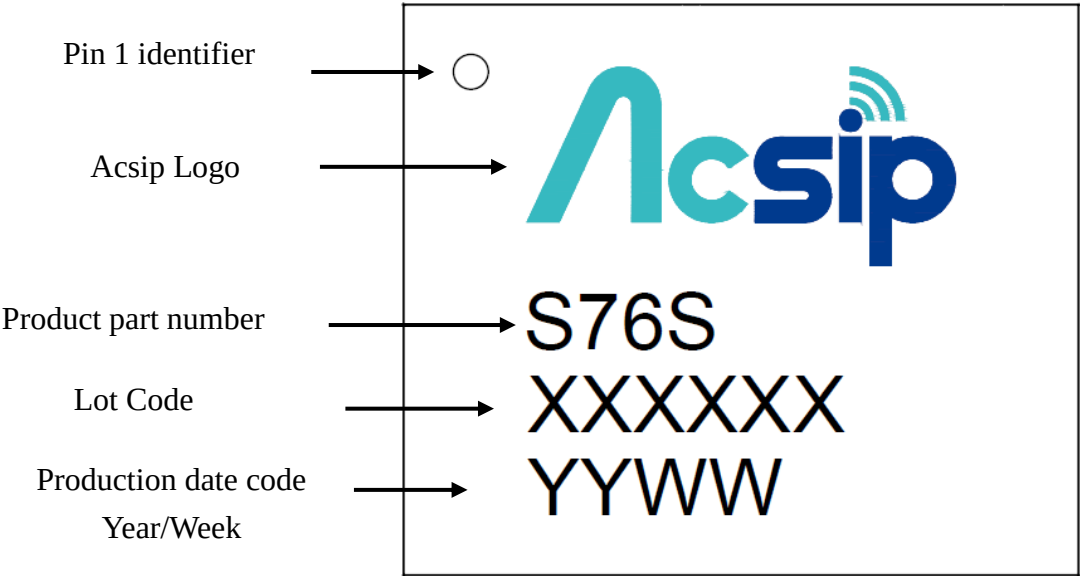
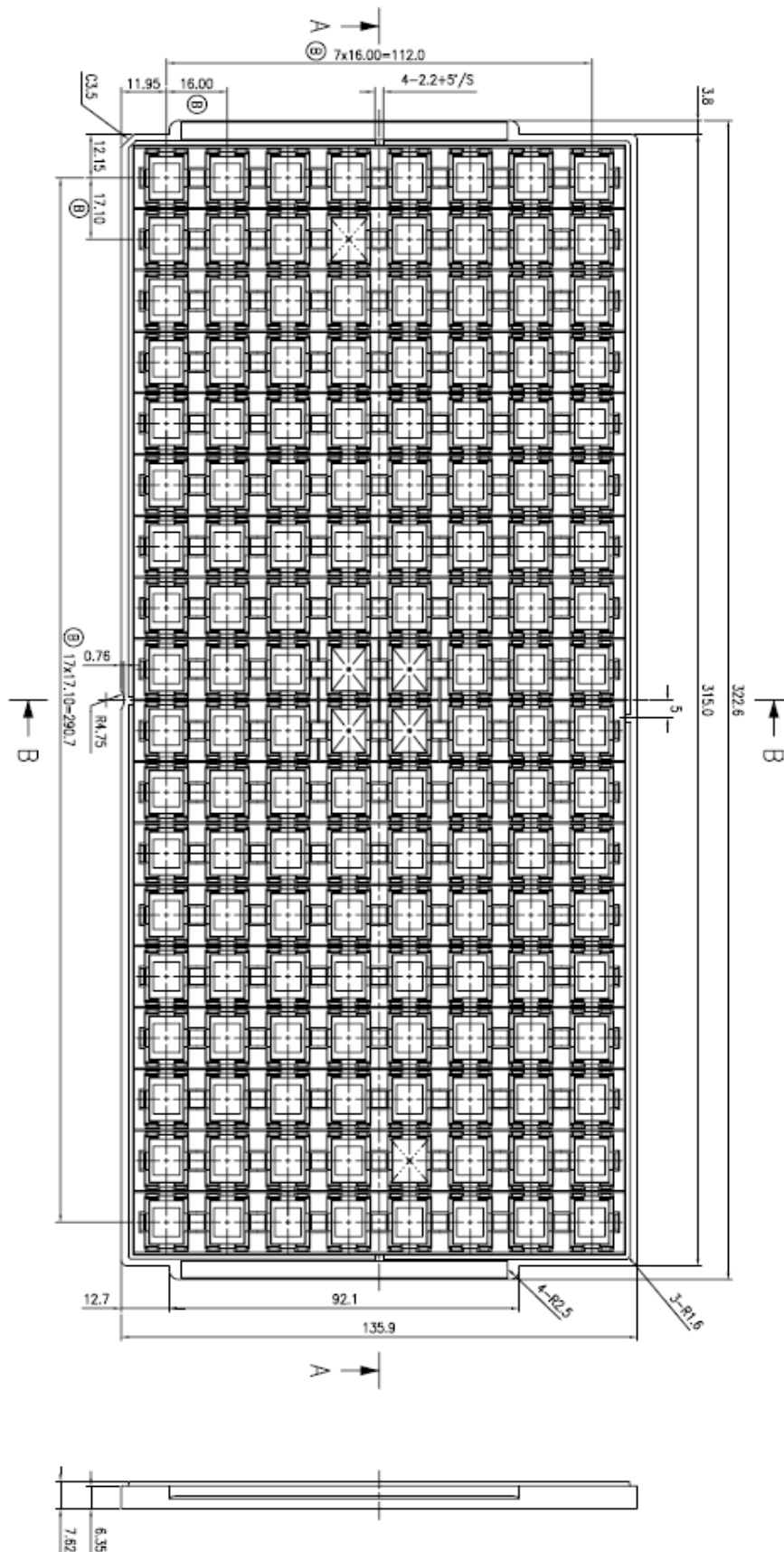


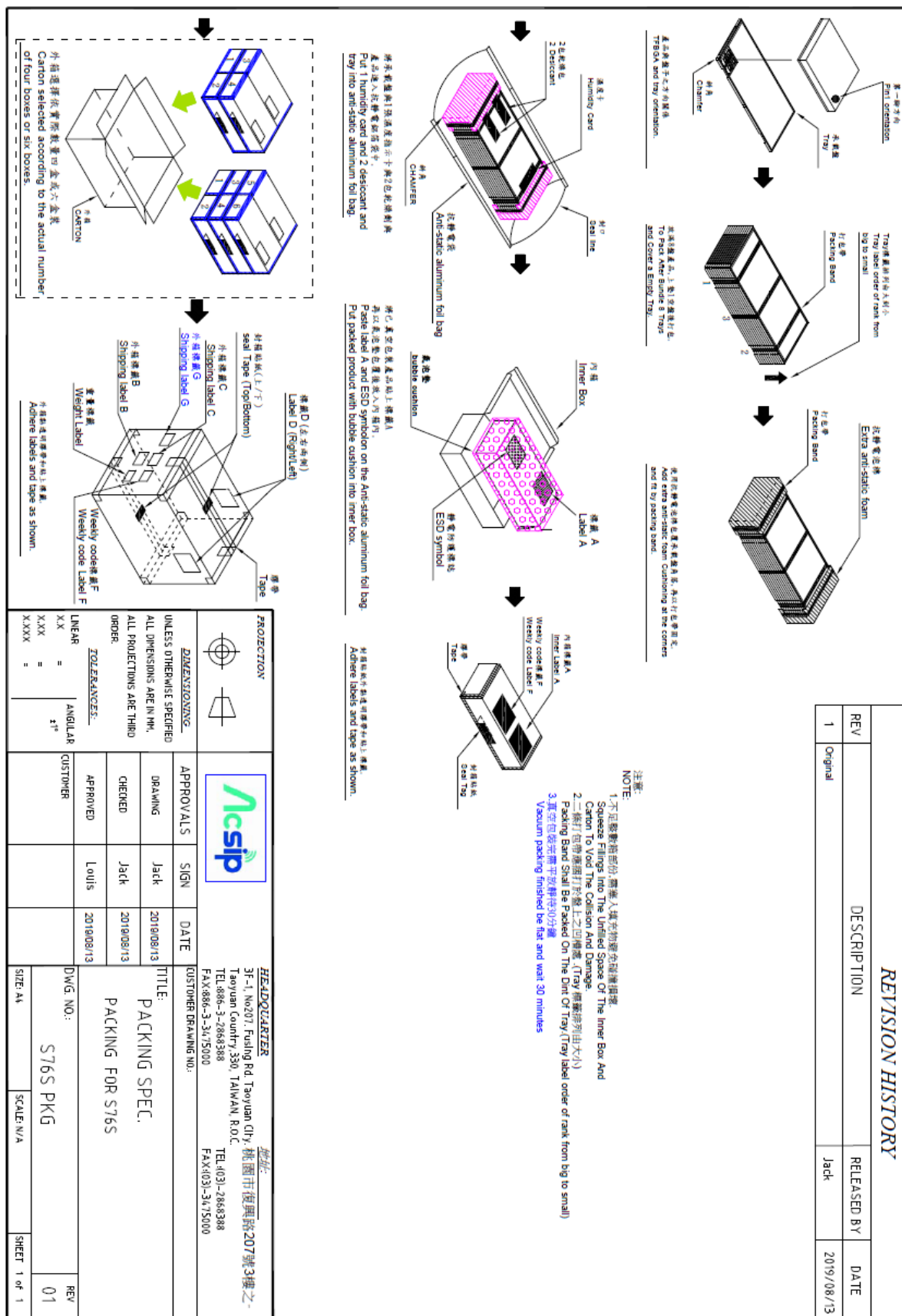
Figure 21Standard Product Marking Diagram- TOP VIEW



7-2. Tray Dimension



7-3. Packing Information



7-4. Humidity Indicator Card



Indicates 指示點:

10%,20%,30,40%,50%,60% relative humidity

10%,20%,30,40%,50%,60% 相對濕度

Color Change 顏色變化:

Brown (Dry) ---> Blue (Wet)

棕色 (乾燥) ---> 藍色 (潮溼)